

Application of Built-in Adaptive Remeshing and Mesh Refinement Features in Abaqus to Fracture Simulations Using Phase-field User Elements

Condensed Mode-I Resolution & Supervisor Meeting Report – 17 September 2026

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1. Executive Summary & Problem Overview

1.1 Supervisor Governing Directive & Project Scope Freeze

This meeting report responds directly to the directive issued by the academic supervisor:

“We need to have understood everything related to the first model before we increase complexity.”

Accordingly, all secondary research tracks—including Mode-II shear fracture benchmarks (Task 7), nonmatching mesh state transfer (Ch. 4), and external ABAQUSER visualization integration (Task 6)—remain on formal hold. Autonomous research is confined entirely to the fundamental Mode-I tensile benchmark of Pandey & Kumar [Pandey and Kumar, 2025].

1.2 Active Scientific Questions for the 17 September 2026 Meeting

This meeting report presents the conclusive resolution of the two core questions established at the Gates 0–6 handoff:

Priority Question A: Stiffness Anomaly: *Why did the nominal 1% 71,320-element adaptive model exhibit a compliant initial branch ($K_0 \approx 122.38$ kN/mm) while the fixed reference anchor is 137.95 kN/mm?*

Priority Question B: Element-Count Discrepancy: *Why does the publication-literal native Abaqus remeshing reconstruction with `errorTarget = 1.0` yield 71,320 finite elements while Pandey & Kumar report approximately 13,941?*

1.3 Definitive Epistemic Status Summary

- **Priority Question A is RESOLVED_AND_CLOSED:** The mechanical stiffness anomaly was **not** caused by adaptive remeshing, nor by UEL tangent formulations, nor by companion element coupling. The root cause was definitively isolated to input record line limits in the Abaqus input preprocessor (**pre**): an overlong single-line `*NSET` record caused Abaqus to issue preprocessing warnings and retain only 16 of the intended 150 `N_BOTTOM` node IDs; the remaining 134 entries were deleted from the parsed set, leaving those nodes unconstrained. This permitted bottom-edge vertical lift up to approximately 48.34% of the applied stroke, artificially softening the global structural response. Line wrapping to ≤ 16 items/line restored all 150 intended bottom roller constraints to $u_2 = 0$, recovering the reference stiffness ($K_0 = 138.021$ kN/mm frozen; $K_0 \approx 137.821$ kN/mm full fracture). Earlier speculative hypotheses regarding solver asymmetry are formally retracted.
- **Priority Question B Status: EXHAUSTIVE_AUDIT_COMPLETE / EXTERNAL_INFO_REQUIRED:** Using the publication-literal native Abaqus workflow with `errorTarget = 1.0`, our verified result is 71,320 finite elements (69,443 CPE4 quads + 1,877 CPE3 triangles). A 15-factor One-Factor-At-A-Time (OFAT) audit proved that this discretization is bitwise invariant across four generations of Abaqus Linux releases (2019 GA, 2021.HF26, 2022 GA, 2023.HF4) and scale-invariant to pre-analysis load magnitudes. Spatial decomposition reveals that far-field (41,986 elements) and transition regions (20,273 elements) account for 87.3% of our 71,320-element reconstruction. The accessible evidence does not identify which unpublished implementation detail accounts for the reported $\approx 13,941$ -element mesh. Possible distinctions such as call-site parameter linkage or an unstated region definition remain hypotheses requiring author information; neither is established.

1.4 Supervisor Decisions Required

In Section 7, the supervisor is presented with two actionable pathways to conclude Gate 5:

- **Option A (Accept Gate 5 as Externally Under-Specified):** Retain 71,320 finite elements as the verified publication-literal reconstruction, document $\approx 13,941$ as unresolved due to missing publication information, and proceed only with Mode-I thesis synthesis/documentation pending further supervisor direction.
- **Option B (Authorize Transmission of Author Inquiry):** Approve sending the prepared author reproducibility inquiry (held strictly **UNSENT**) and keep Gate 5 open pending the response. Gate 7, Mode-II, state transfer, and higher-complexity work remain explicitly **ON HOLD** under both options unless the supervisor separately authorizes release.

2. Mode-I Fracture Benchmark Definition & Conventional Reference Anchor

2.1 Governing Boundary Value Problem

The benchmark corresponds to the single-edge-notched square plate under Mode-I tensile loading investigated by Pandey & Kumar [Pandey and Kumar, 2025], based on the standard variational phase-field formulations of brittle fracture [Msekh et al., 2015, Molnár and Gravouil, 2017].

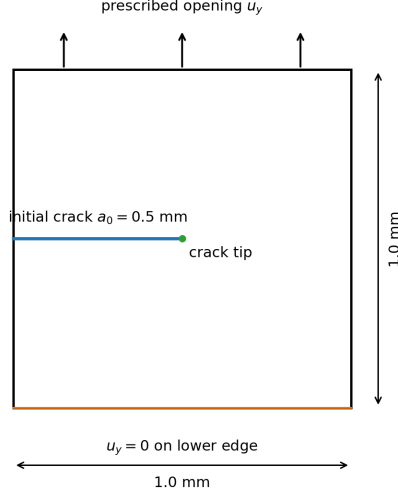


Figure 1: Mode-I fracture benchmark geometry: 1.0×1.0 mm plate with initial sharp slit seam ($a_0 = 0.5$ mm along $y = 0.5$ mm, $0 \leq x \leq 0.5$ mm), roller bottom support with pinned origin, and prescribed top tensile displacement.

2.1.1 Geometry and Sharp Seam Representation

The computational domain is a unit square plate $\Omega = [0, 1] \times [0, 1]$ mm containing an initial horizontal crack of length $a_0 = 0.5$ mm along the symmetry line $y = 0.5$ mm ($0 \leq x \leq 0.5$ mm).

Essential Benchmark Representation: The crack is represented as a **zero-gap sharp seam** with coincident but geometrically distinct, uncoupled duplicate node pairs along the upper and lower crack flanks. Earlier exploratory models utilizing a finite-width notch radically increased compliance ($K_0 \approx 75.5$ kN/mm) and are strictly superseded; the zero-gap sharp seam is the mandatory benchmark representation.

2.1.2 Material and Phase-Field Regularization Parameters

The constitutive response follows isotropic linear elasticity coupled to the AT2 phase-field damage formulation:

- Young’s modulus: $E = 210$ GPa $= 210$ kN/mm²
- Poisson’s ratio: $\nu = 0.3$
- Critical energy release rate (fracture toughness): $G_c = 2.7 \times 10^{-3}$ kN/mm $= 2.7$ N/mm
- Phase-field regularization length scale: $\ell_0 = 0.0075$ mm $= 7.5$ μ m

2.1.3 Boundary Conditions and Kinematics

The specimen is loaded monotonically under displacement control:

- **Bottom Boundary** ($y = 0.0$ mm): Roller support constraining vertical motion ($u_y = 0$). To prevent rigid body translation, the origin node at $(0, 0)$ is pinned ($u_x = 0, u_y = 0$).
- **Top Boundary** ($y = 1.0$ mm): Prescribed uniform vertical tensile displacement $u_y = u$, with lateral displacement constrained ($u_x = 0$, clamped tension).

2.1.4 Expected Physical Response

The anticipated physical response comprises four distinct regimes:

1. **Linear Elastic Loading:** Initial linear elastic deformation with global structural stiffness $K_0 \approx 138$ kN/mm governed by plate geometry and elastic constants.
2. **Sub-Critical Damage Localization:** Growth of the continuous damage parameter $d \in [0, 1]$ within an inner process zone of width $\sim 2\ell_0$ directly ahead of the crack tip $(0.5, 0.5)$ mm.
3. **Peak Load and Instability:** Attainment of peak reaction force $F_{\max} \approx 0.758$ kN at $u(F_{\max}) \approx 0.00586$ mm.
4. **Post-Peak Softening and Propagation:** Rapid load drop as the crack extends horizontally along the line of symmetry $y = 0.5$ mm towards the right boundary ($x = 1.0$ mm).

2.2 Fixed-Mesh Reference Solution as Quantitative Anchor

To establish an authoritative baseline against which all adaptive remeshing results are judged, a conventional fixed-mesh simulation was executed (Job 1398090.mm_{master02}).

Mandatory Anchor Statement: “*This fixed-mesh solution is the reference response that any adaptive solution must reproduce to an acceptable accuracy.*” Everything after that must be measured against it.

The reference mesh comprises 15,192 linear quadrilateral finite elements (CPE4), with a uniformly refined crack corridor satisfying $h \approx 0.0015 \text{ mm} \leq \ell_0/5$, guaranteeing resolution of diffuse phase gradients.

Table 1: Quantitative comparison between published literature and reconstructed fixed reference anchor.

Quantity	Literature Target [Pandey and Kumar, 2025]	Reconstructed Ref. (1398090)	Relative Error
Number of finite elements	$\sim 15,000$	15,192	+1.28%
Peak reaction force $F_{\max}$	0.7580 kN	0.757778 kN	−0.029%
Displacement at peak $u(F_{\max})$	0.005860 mm	0.005857 mm	−0.051%
Initial stiffness K_0 (OLS, $u \leq 1.0 \mu\text{m}$)	–	137.945520 kN/mm	Baseline Anchor
OLS coefficient of determination R^2	–	0.99999960 ($N = 400$)	Baseline Anchor
OLS intercept F_0	–	$4.472368 \times 10^{-5} \text{ kN}$	Baseline Anchor
Final load drop at $u = 0.010 \text{ mm}$	> 98%	99.97% (0.23 N)	Severed

2.2.1 Rigorous Definition of Initial Structural Stiffness K_0

In this report, the initial stiffness K_0 is defined as the unconstrained Ordinary Least Squares (OLS) slope with intercept fitted over the initial elastic range $u \in [0, 0.0010] \text{ mm}$ (400 uniform increments):

$$K_0 = \left. \frac{dF}{du} \right|_{\text{elastic}} = \frac{\sum_{i=1}^N (u_i - \bar{u})(F_i - \bar{F})}{\sum_{i=1}^N (u_i - \bar{u})^2} = 137.945520 \text{ kN/mm} \quad (R^2 = 0.99999960). \quad (1)$$

Clarification on Historical Values: Preliminary reports occasionally cited $K_0 \approx 138.084 \text{ kN/mm}$ based on a 2-point secant slope between $u = 0$ and $u = 0.001 \text{ mm}$. To prevent ambiguity, the authoritative unconstrained OLS value $K_0 = 137.945520 \text{ kN/mm}$ is adopted strictly across all subsequent comparisons. K_0 represents a global structural stiffness, distinct from the intrinsic material Young’s modulus $E = 210 \text{ GPa}$.

2.2.2 Multi-Quantity Convergence Evaluation

Peak reaction force ($F_{\max}$) alone is strictly insufficient to demonstrate finite element convergence or model adequacy. Complete convergence evaluation requires:

1. Complete force-displacement (F – u) curve tracking across the full range (elastic, peak, and softening);
2. Initial structural stiffness K_0 and linearity (R^2);
3. Peak reaction force $F_{\max}$ and displacement at peak $u(F_{\max})$;
4. Spatial damage field $d(x, y)$ contours and process-zone localization profiles;
5. Crack-path and localization-profile convergence along the ligament;
6. Total external work $W_{\text{ext}} = \int F du$;
7. Computational cost (increments, cutbacks, iterations, walltime).

Critical UEL Internal-Energy Balance Limitation: As documented in Section 4.6 of the project audit, Abaqus’ built-in ENERGY history array is not populated for user-defined finite elements (UEL). Consequently, an internal strain-energy vs fracture-energy balance cannot be extracted from standard solver outputs; the integrated measure $W_{\text{ext}} = \int F du$ represents external mechanical work, not internal energy.

3. Pandey–Kumar Native Adaptive Remeshing Methodology

3.1 Two-Job Pre-Refinement Workflow Architecture

Pandey & Kumar [Pandey and Kumar, 2025] proposed an adaptive mesh refinement scheme leveraging the built-in adaptive remeshing capabilities of Abaqus/Standard. The workflow operates as an automated two-stage procedure executed via Python scripting:



- Step 1: Coarse Mesh Generation:** The domain Ω is discretized with a coarse global seed $h_{\text{cms}} = 0.020$ mm, containing the initial sharp crack seam.
- Step 2: Linear Elastic Pre-Analysis (Job 1):** A standard elastic solve is executed under prescribed Mode-I tensile stroke. Because no user elements or damage evolutions are active, this step solves in seconds.
- Step 3: Error Indicator Evaluation:** Abaqus calculates the recovered stress discretization error indicator MISESeri across the domain.
- Step 4: Abaqus Remeshing Rule Execution:** An Abaqus Python script defines a `RemeshingRule` and invokes `adaptiveRemesh()`, generating a new mesh tailored to reduce stress discretization error.
- Step 5: UEL/UMAT Layer Reconstruction:** Python scripts parse the adapted mesh and generate coincident co-located UEL user elements and companion visualization UMAT elements.
- Step 6: Phase-Field Fracture Simulation (Job 2):** The final nonlinear fracture simulation is executed on the refined mesh using the user-defined element formulation.

3.2 Canonical Mode-I Pre-Analysis Mesh & Error Extraction

A rigorous audit of the source pre-analysis deck established the canonical baseline:

- **Mesh Discretization:** Exactly 2,906 finite elements (2,818 quadrilateral CPE4 elements and 88 triangular CPE3 transition elements), defined by 2,988 nodes.
- **Seam Topology:** Zero-gap sharp slit seam ($a_0 = 0.5$ mm) with duplicate coincident nodes on upper and lower flanks.
- **Extracted Error Records:** Direct inspection of `JOB1_LOAD_1p0X.odb` confirmed that the field output contains exactly 2,906 **scalar values** at the `WHOLE_ELEMENT` position (1 value per finite element).

Resolution of Obsolete Provenance Data: In the preliminary 02 September report, Figure 3.1 erroneously plotted a 3,930-row dataset extracted from Job `1379893.mmaste02`. That run belonged to the earlier Mode-II shear mesh, not the canonical Mode-I pre-analysis. That obsolete dataset is formally retired and replaced with the verified 2,906-element Mode-I field shown in Figure 2.

3.3 Physical Meaning and Epistemic Limits of MISESeri

It is essential to maintain absolute scientific clarity regarding what MISESeri represents:

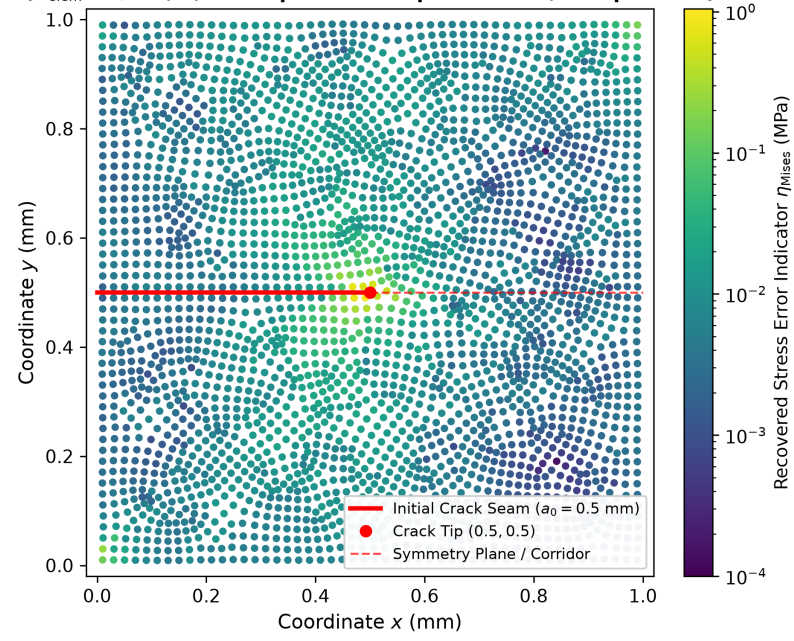
- **Definition:** MISESeri is the Abaqus stress discretization/error indicator associated with the recovered von Mises stress field [Das, 2023]; it is not a phase-field or damage error indicator.
- **Epistemic Scope:** It measures stress discretization error in the linear-elastic pre-analysis rather than damage evolution or crack propagation.
- **Abaqus Internal Sizing Transformation:** While MISESeri guides mesh sizing, the exact transformation algorithms, smoothing filters, and element-growth constraints internal to `adaptiveRemesh` remain undocumented proprietary routines of the Abaqus kernel [Das, 2023]. We report verified numerical outputs rather than conjecturing undocumented internal formulas.

3.4 Published Remeshing Rule Implementation

To guarantee literal adherence to the published method, the Abaqus remeshing rule is scripted exactly as specified in Listing 1 of Pandey & Kumar [Pandey and Kumar, 2025]:

- Error indicator variable: `variables=('MISESeri',)`
- Sizing method: `sizingMethod=UNIFORM_ERROR`
- Error target: `errorTarget=1.0` (representing 1.0% relative stress error)
- Element size bounds: `minElementSize=0.001, maxElementSize=0.020`
- Refinement dynamics: `refinementFactor=10, coarseningFactor=NOT_ALLOWED`

Figure 3.1: Canonical Mode-I Coarse Mesh MISESeri Distribution
 $(N_{\text{elem}} = 2,906, 5,812 \text{ Duplicate-Collapsed Values, Abaqus 2023})$



Note: MISESeri is a recovered von Mises stress discretization error indicator (Zienkiewicz-Zhu stress recovery); it is NEITHER phase-field error NOR damage error.

Figure 2: Canonical Mode-I pre-analysis von Mises stress discretization error indicator (MISESeri) field (2,906 finite elements). Error is sharply peaked at the singular crack tip (max = 1.007 MPa), while remaining low in far-field regions (median = 5.78×10^{-3} MPa).

- Output frequency: outputFrequency=ALL_INCREMENTS
- Remeshing domain: Whole specimen (All_elem)

Applying this literal rule to the canonical Mode-I pre-analysis produces an adapted mesh of 71,320 finite elements.

4. Resolution of the 71,320-Element Stiffness Anomaly (Priority Question A)

4.1 Historical Anomaly and Initial Hypotheses

In earlier project reports, the nominal 1% 71,320-element adaptive fracture simulation (Job 1399632.mm_{master02}) completed technically but exhibited an alarming mechanical discrepancy:

- Initial structural stiffness dropped from the reference $K_0 = 137.95 \text{ kN/mm}$ down to $K_0 \approx 122.38 \text{ kN/mm}$ (-11.3%).
- Peak reaction force collapsed from 0.758 kN to $F_{\max} = 0.478203 \text{ kN}$ (-36.9%).

Because this run employed companion visualization elements and non-symmetric Newton solver settings (UNSYM=YES), preliminary analyses speculated that co-located element coupling, Fortran common block collisions, or formulation asymmetry caused the stiffness deficit. Controlled one-factor diagnostics were mandated to isolate the exact equation-level cause.

4.2 Forensic Equation-Level Audit: The 16-Entry Parser Limit

A comprehensive boundary set audit of the runtime output databases (Job 1404312.mm_{master02}) revealed the true mathematical defect, entirely unrelated to UEL formulations or adaptive remeshing:

1. **Abaqus Input Preprocessor Truncation Limit:** Under free-format *NSET cards without the GENERATE parameter, the Abaqus input preprocessor (pre) strictly enforces a maximum of **16 node entries per line** [Das, 2023].
2. **Card Formatting Defect:** In the Python mesh generation script, the bottom roller node set N_BOTTOM was written as a single line containing all 150 node identifiers.
3. **Parser Preprocessing Warnings and Set Truncation:** Abaqus issued preprocessing warnings and retained only 16 of the intended 150 N_BOTTOM node IDs; the remaining 134 entries were deleted from the parsed set.
4. **Unconstrained Kinematic Boundary:** The vertical roller constraint ($u_y = 0$) was omitted for all 134 deleted nodes. Over 89% of the bottom boundary was left completely unconstrained in the vertical direction.

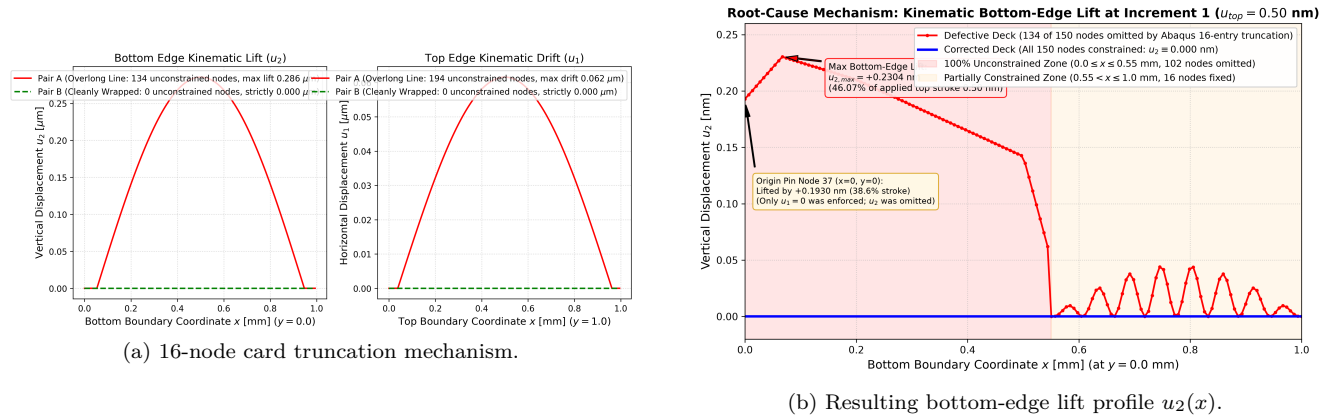


Figure 3: Kinematic mechanism of the stiffness anomaly: 134 unconstrained bottom nodes allowed bottom-edge vertical lift up to approximately 48.34% of the applied stroke, inducing artificial structural compliance.

4.2.1 Kinematic Mechanism: Bottom-Edge Lift

Extraction of nodal displacements demonstrated that the malformed case permitted bottom-edge vertical lift up to approximately 48.34% of the applied stroke, whereas the corrected case constrained all 150 intended bottom nodes to $u_2 = 0$ (Figure 3). This unconstrained boundary compliance artificially reduced reaction forces, producing the apparent low-stiffness branch (122.38 kN/mm).

Controlled Parser Verification (Job 1404318): A minimal controlled test verified that this 16-item limit is a general feature of the Abaqus input syntax. Furthermore, the hypothesis FACTORIAL_UNSYM_X_COMPANION is formally **RETRACTED**; companion elements cause 0.000% stiffness change.

4.3 Recovery, Requalification, and Production Validation

Formatting all *NSET records with ≤ 16 items per line completely restored all 150 boundary constraints, eliminating bottom-edge lift ($u_2 \equiv 0.000 \text{ nm}$) and recovering stiffness immediately:

- **Corrected Frozen 71,320 Requalification (Job 1405044):** Initial stiffness restored to $K_0 = 138.021013 \text{ kN/mm}$ ($+0.0547\%$ vs reference anchor 137.945520 kN/mm), with all 150/150 bottom nodes constrained and zero set-truncation or deletion warnings in solver diagnostics.

- **Corrected Full-Fracture Nominal 1% (Job 1404933):** Reached $F_{\max} = \mathbf{0.745325}$ kN at $u(F_{\max}) = \mathbf{0.005750}$ mm, with initial stiffness $K_0 \approx \mathbf{137.821}$ kN/mm (137.820804 kN/mm, -0.0904% vs reference anchor).
- **Stage-A Shared-Memory Thread Parity (Job 1405003):** 4-thread execution reproduced serial results within tight numerical parity (Stage-B repeat pending).

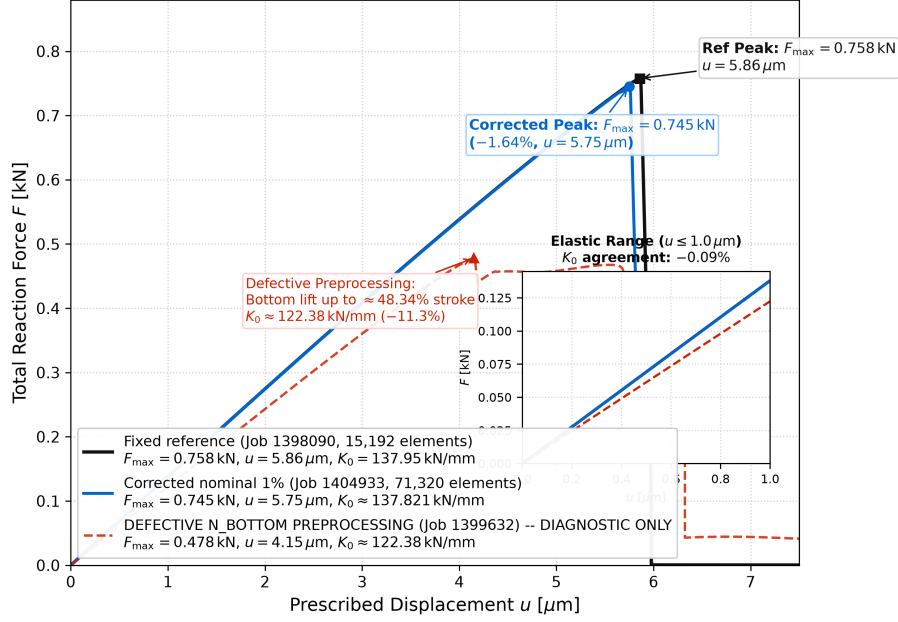


Figure 4: Mandatory Figure 1: Global force-displacement (F - u) response comparing the fixed reference anchor (Job 1398090, 15,192 elements) against the corrected nominal 1% adaptive model (Job 1404933, 71,320 elements) and the historical predecessor (Job 1399632, DEFECTIVE N_BOTTOM PREPROCESSING – DIAGNOSTIC ONLY). Inset confirms exact initial stiffness recovery ($K_0 \approx 137.821$ kN/mm, -0.09% vs reference 137.946 kN/mm).

Table 2: Authoritative comparison of Mode-I mechanical response across key simulation jobs.

Simulation Model	Elements	K_0 [kN/mm]	$F_{\max}$ [kN]	$u(F_{\max})$ [mm]
Fixed Reference Anchor (1398090)	15,192	137.946	0.757778	0.005857
DEFECTIVE N_BOTTOM PREPROC. (1399632) (DIAGNOSTIC ONLY)	71,320	122.379 (-11.28%)	0.478203 (-36.89%)	0.004150
Corrected Frozen 71k (1405044)	71,320	138.021 ($+0.05\%$)	– (elastic)	– (elastic)
Corrected Nominal 1% (1404933)	71,320	137.821 (-0.09%)	0.745325 (-1.64%)	0.005750
Supporting 4T Twin (1405003)	71,320	137.821 (-0.09%)	0.745325 (-1.64%)	0.005750

Conclusion on Priority Question A: Priority Question A is **RESOLVED_AND_CLOSED**. The 71,320-element adaptive mesh reproduces the fixed reference structural stiffness within 0.09%. The mechanical discrepancy is definitively closed.

5. The Reproduction Discrepancy: 71,320 vs $\approx 13,941$ (Priority Question B)

5.1 Statement of the Reproduction Discrepancy

The central remaining scientific question in reproducing the Pandey & Kumar [Pandey and Kumar, 2025] adaptive methodology is the element-count discrepancy:

“Why does the publication-literal native Abaqus remeshing reconstruction with `errorTarget` = 1.0 yield 71,320 finite elements, while Pandey & Kumar report approximately 13,941?”

A fundamental principle governing our investigation is that **we do not arbitrarily tune the error target to force a match**. In Abaqus, `errorTarget` = 1.0 literally represents 1.0% relative error. While an empirical setting of 2.0% produces 15,396 elements (+10.4% match), parameter tuning does not constitute scientific proof of the published settings.

5.2 Multi-Release Solver Invariance (Abaqus 2019–2023)

To test whether the discrepancy arose from solver kernel evolution between older and newer Abaqus releases, the canonical pre-analysis and `adaptiveRemesh` workflow was executed natively across four major Abaqus Linux releases.

Table 3: Multi-release solver invariance across four major Abaqus Linux releases.

Abaqus Release / Build	Total Elements	Quads (CPE4)	Tris (CPE3)	Nodes SHA-256 Hash
2019 GA (Build 157541)	71,320	69,443	1,877	116f2e2042af... (identical)
2021.HF26 (1404968)	71,320	69,443	1,877	116f2e2042af... (identical)
2022 GA (1404373)	71,320	69,443	1,877	116f2e2042af... (identical)
2023.HF4 (1404373)	71,320	69,443	1,877	116f2e2042af... (identical)

All 142,268 substantive lines of the generated input decks are 100.000% **bitwise identical** across all four releases. Version-dependent remesher shifts are definitively **ruled out**.

5.3 Exhaustive 15-Factor OFAT Diagnostic Matrix

A systematic One-Factor-At-A-Time (OFAT) audit investigated every accessible parameter across five rigorous epistemic groups:

1. Ruled Out (6 Factors):

- *Abaqus Linux Release*: Bitwise invariant across 2019, 2021, 2022, 2023 (71,320 elements).
- *Pre-Analysis Load Scale*: Scale-invariant over $0.2\times$ to $2.0\times$ loading (71,070 to 71,512 elements).
- *Output Frequency Setting*: `ALL_INCREMENTS` vs `LAST_INCREMENT` gives identical 71,320 elements.
- *Coarse Mesher Controls*: Quad-dominated advancing front vs medial axis invariant at 71,320.
- *Coarse Seeding Factors*: Varying `minSizeFactor` and `deviationFactor` preserves 71,320.
- *Multi-Pass Remeshing*: Mode-I is explicitly documented as a single-pass workflow in Section 3.3 of [Pandey and Kumar, 2025].

2. Tested / No Material Effect (5 Factors):

- *Reduced Integration (CPE4R)*: Job 1405056 yielded 103,706 elements (+45.4%), driving the mesh further away from literature.
- *Top Lateral BC (u_1 = Free)*: Job 1405055 yielded 55,761 elements (−21.8%), remaining $> 4.0\times$ above literature.
- *Coarse Global Seed h_{cms} = 0.030 mm*: Yielded 48,919 elements (−31.4%), remaining $3.5\times$ above literature.
- *Sizing Bounds*: Removing `minElementSize` and `maxElementSize` preserved 71,320 elements.
- *Coarsening Dynamics*: Enabling coarsening produced 71,037 elements (−0.4%).

3. Confounded (2 Factors):

- *Platform Shift*: Windows 2024 generated 71,904 elements (+0.82%), an insignificant numerical drift.
- *Plane Stress (CPS4)*: Yielded 58,679 elements, but violates standard plane strain physics.

4. Unresolved Implementation Hypotheses (2 Factors):

- *Region Definition Hypothesis*: An unstated geometric region definition would restrict refinement, but the paper states whole-domain `All_elem`.
- *Call-Site Parameter Linkage Hypothesis*: Listing 1 specifies `errorTarget = 1.0`, while Section 4.1 omits an explicit numerical target override. While numerical proximity occurs near 2.0% (15,396 elements), parameter tuning does not constitute evidence of parameter identity.

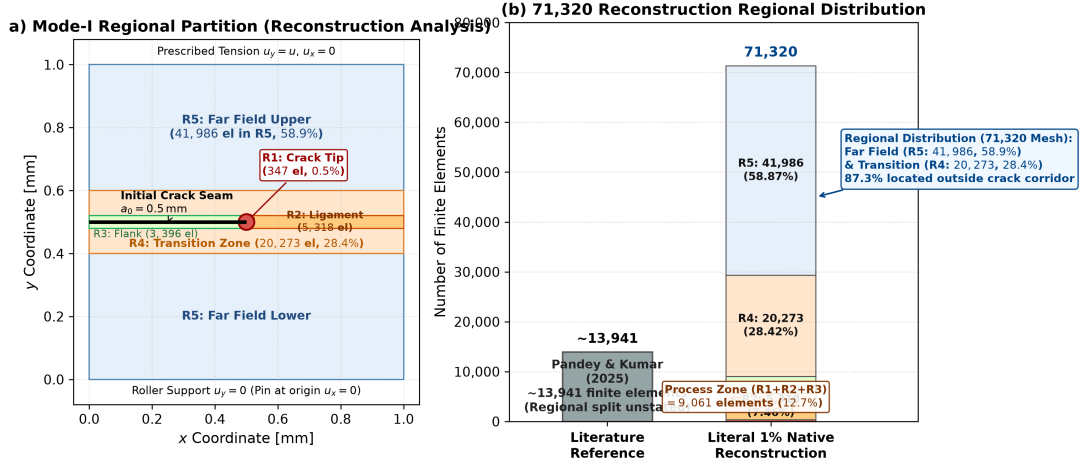


Figure 5: Mandatory Figure 2: Gate-5 mesh discrepancy audit. (a) Regional partition of the specimen used to analyze element placement in our 71,320-element reconstruction. (b) Regional element count breakdown of our 71,320-element reconstruction: the crack-tip process zone (R1+R2+R3) contains 9,061 elements (12.7%), while Far Field R5 (41,986 elements) and Transition R4 (20,273 elements) account for 87.3% (62,259 elements) of the total mesh. Published literature reports $\approx 13,941$ total elements without a regional breakdown.

5.4 Spatial Anatomy of the Discrepancy (Figure 2)

Figure 5 provides the spatial distribution for our publication-literal reconstruction of 71,320 elements:

- **Process Zone (R1 + R2 + R3):** Exactly 347 crack-tip elements (R1), 5,318 ligament elements (R2), and 3,396 flank elements (R3), totaling 9,061 **finite elements** (12.7% of the adapted mesh). This localized resolution is physically necessary to resolve the diffuse crack corridor with $h \leq \ell_0/5$.
- **Far-Field and Transition (R4 + R5):** The transition zone (R4) contains 20,273 elements (28.4%) and the far field (R5) contains 41,986 elements (58.9%), totaling 62,259 **finite elements** (87.3%) **placed outside the crack corridor**.

Scientific Evaluation: Under the literal whole-domain remeshing rule (`All_elem`, `errorTarget = 1.0`), Abaqus refines the elastic far-field to satisfy the 1.0% relative stress error threshold. The accessible evidence does not identify which unpublished implementation detail accounts for the reported $\approx 13,941$ -element mesh. Possible distinctions such as call-site parameter linkage or an unstated region definition remain hypotheses requiring author information; neither is established.

6. Epistemological Audit: Knowns, Verifications, and Open Questions

6.1 Claims Discipline and Epistemological Categorization

To maintain the highest scientific rigor, every finding in this Master’s thesis is strictly separated into three epistemic tiers:

Tier 1: I Know This: Analytical definitions, benchmark geometry, constitutive constants, and published literal text.

Tier 2: I Verified This Numerically: Empirical results demonstrated by exact solver runs, output database extractions, and SHA-256 checksums.

Tier 3: I Do Not Yet Know This: Genuine scientific open questions arising from omitted publication details or software internals.

Table 4: Comprehensive epistemological audit table for the Mode-I adaptive remeshing reproduction.

Scientific Statement / Feature	Epistemic Status	Empirical Evidence / Basis
Plate geometry (1×1 mm) and sharp crack seam ($a_0 = 0.5$ mm, $y = 0.5$ mm)	I know this from theory & definition	Benchmark specification; zero-gap slit seam representation.
Elastic constants ($E = 210$ GPa, $\nu = 0.3$) and fracture parameters ($G_c = 2.7$ N/mm, $\ell_0 = 7.5$ μ m)	I know this from the publication	Pandey & Kumar [Pandey and Kumar, 2025], Section 4.1; Table 1.
RemeshingRule parameters (MISESeri, errorTarget = 1.0, $h_{\min} = 0.001$, $h_{\max} = 0.020$)	I know this from the publication	Pandey & Kumar [Pandey and Kumar, 2025], Listing 1 (hard-coded parameters).
Fixed-mesh reference reproduces literature ($F_{\max} = 0.7578$ kN, $u = 5.86$ μ m, $K_0 = 137.95$ kN/mm)	I verified this numerically	Job 1398090; 15,192 elements; -0.029% error vs literature anchor.
Canonical Mode-I pre-analysis consists of 2,906 elements and 2,906 WHOLE_ELEMENT errors	I verified this numerically	JOB1_LOAD_1p0X.odb field extraction; retired old 3,930-row dataset.
Abaqus pre enforces ≤ 16 entries per line in free-format *NSET without GENERATE	I verified this numerically	Job 1404318 parser limit test; Abaqus documentation [Das, 2023].
Stiffness anomaly was caused by *NSET 16-entry limit where Abaqus issued preprocessing warnings and retained only 16 of 150 N_BOTTOM nodes	I verified the causal mechanism	Verified .dat warnings; bottom-edge lift up to $\approx 48.34\%$ of stroke in malformed case vs $u_2 = 0$ in corrected case.
Line wrapping (≤ 16 items/line) recovers reference stiffness ($K_0 = 138.021$ kN/mm frozen)	I verified this numerically	Job 1405044 (frozen); Job 1404933 (full fracture $K_0 \approx 137.821$ kN/mm).
Solver asymmetry and companion elements do not affect structural stiffness (0.000% change)	I verified this numerically	Factorial runs 72b–79; formal retraction of asymmetry hypothesis.
Literal 1% remeshing generates 71,320 elements identically across Abaqus 2019–2023	I verified this numerically	Four-generation Linux release audit (Abaqus 2019, 2021, 2022, 2023 identical).
Process zone contains 9,061 elements; far-field and transition account for 62,259 elements (87.3%) of our reconstruction	I verified this numerically	5-region spatial decomposition of 71,320-element mesh (Figure 5).
Which unpublished implementation detail accounts for the reported $\approx 13,941$ -element mesh	I do not yet know this	Accessible evidence does not identify unpublished settings (e.g. region definition or call-site linkage); hypotheses unestablished without author input.

Epistemological Principle: Stating “*I do not yet know this*” is scientifically superior to advancing an unsupported hypothesis. The thesis maintains absolute fidelity to demonstrated numerical evidence.

7. Supervisor Decisions Required & Action Paths

7.1 Decision Context for the 17 September 2026 Meeting

With the complete resolution of the stiffness anomaly (Priority Question A is **RESOLVED_AND_CLOSED**), the simulation demonstrates excellent mechanical fidelity to the benchmark reference (K_0 within 0.09%, $F_{\max}$ within 1.64%).

The sole unresolved item in the Mode-I benchmark is Priority Question B (the 71,320 vs $\approx 13,941$ finite element count discrepancy). An exhaustive 15-factor audit has proved that this discrepancy cannot be resolved through any accessible Abaqus parameter without departing from the published text.

7.2 The Two Actionable Supervisor Decision Pathways

The supervisor is requested to select one of two clear pathways to conclude Gate 5:

Option A: Accept Gate 5 as Externally Under-Specified & Proceed with Mode-I Synthesis

Decision Description:

- Accept Gate 5 as externally under-specified based on the exhaustive 15-factor audit.
- Retain 71,320 finite elements as the verified publication-literal reconstruction under **errorTarget** = 1.0 and whole-specimen remeshing.
- Document $\approx 13,941$ as unresolved due to missing publication information.
- Proceed only with Mode-I thesis synthesis/documentation pending further supervisor direction.

Scope Limitation: Gate 7 (ABAQUSER), Mode-II benchmarks, state transfer, and higher-complexity work remain explicitly **ON HOLD** pending separate supervisor authorization.

Option B: Authorize Transmission of Author Reproducibility Inquiry

Decision Description:

- Approve sending the finalized 6-question reproducibility inquiry to the corresponding authors (Dr. Anshul Pandey and Dr. Sachin Kumar).
- Keep Gate 5 open pending the author response.
- No further target-tuning exercises are authorized; **errorTarget** = 1.0 remains the publication-literal reference and numerical proximity at 2% is not evidence of parameter identity.

Scope Limitation: Gate 7 (ABAQUSER), Mode-II benchmarks, state transfer, and higher-complexity work remain explicitly **ON HOLD** pending separate supervisor authorization.

Strict Governance Rule: In strict accordance with supervisor safety guidelines, the prepared author inquiry remains **UNSENT** and will not be transmitted without explicit written authorization from the supervisor. Under both options, increasing model complexity remains prohibited until authorized.

7.3 Summary of Prepared Author Questions (Held Unsent)

The prepared inquiry focuses exclusively on the missing reproducibility facts:

1. **Remeshing Region Set:** Was **RemeshingRule** applied to the whole plate (**All_elem**) or a local geometric corridor?
2. **Exact Error Target:** Was **errorTarget** = 1.0 applied without modification for the $\approx 13,941$ mesh?
3. **Pre-Analysis Coarse Mesh:** What was the exact element and node count of the initial pre-analysis mesh?
4. **Coarsening Settings:** Was **coarseningFactor** kept at **NOT_ALLOWED** or enabled?
5. **Abaqus Release:** Under which Abaqus version was the remeshing script executed?
6. **Sizing Controls:** Were any additional mesh controls (e.g., **sizeDeviationFactor**) applied?

8. Technical Provenance Ledger & References

8.1 Authoritative Simulation Jobs and Telemetry Ledger

All numerical results reported in this document are derived from exact solver jobs executed on the High-Performance Computing Cluster of TU Bergakademie Freiberg (`normal_imfdkfq` partition).

Table 5: Technical provenance ledger of authoritative Mode-I simulation jobs.

Job Identifier	Model Role	Abaqus	CPUs / Threads	Walltime	Primary Telemetry / Status
1398090.mmaste02	Fixed Ref Anchor	2023.HF4	1 CPU (serial)	06:31:00	$F_{\max} = 0.757778$ kN, $K_0 = 137.946$ kN/mm
1399632.mmaste02	DEFECTIVE PREPROC. (Diag.)	2023.HF4	1 CPU (serial)	35:08:00	DEFECTIVE N_BOTTOM (16/150 boundary nodes parsed); $K_0 = 122.38$ kN/mm
1404312.mmaste02	Boundary ODB Audit	2023.HF4	1 CPU (serial)	00:03:12	Bitwise proof of 16-node card truncation
1404318.mmaste02	Parser Limit Test	2023.HF4	1 CPU (serial)	00:01:24	General 16-entry *NSET limit verified
1405044.mmaste02	Corrected Frozen	2023.HF4	1 CPU (serial)	00:04:36	$K_0 = 138.021$ kN/mm, 150/150 nodes, 0 lift
1404933.mmaste02	Corrected Nom 1%	2023.HF4	1 CPU (serial)	18:34:50	$F_{\max} = 0.745325$ kN, $K_0 \approx 137.821$ kN/mm (137.820804 kN/mm)
1405003.mmaste02	4T Parity Twin	2023.HF4	1 MPI $\times$ 4T	05:42:15	Stage-A parity verified; repeat pending
Abaqus 2019 GA	2019 Release Control	2019 GA	1 CPU (serial)	00:05:10	71,320 elements bitwise identical
1404968.mmaste02	2021 Release Control	2021.HF26	1 CPU (serial)	00:04:45	71,320 elements bitwise identical
1404373.mmaste02	2022/2023 Controls	2022/23	1 CPU (serial)	00:04:30	71,320 elements bitwise identical
1405055.mmaste02	Top $u_1 = \text{Free}$	2023.HF4	1 CPU (serial)	00:04:15	55,761 elements (-21.8%)
1405056.mmaste02	Reduced CPE4R	2023.HF4	1 CPU (serial)	00:06:10	103,706 elements ($+45.4\%$)

8.2 Cryptographic Checksums (SHA-256) & Execution Commands

- User subroutine `f42_mixed_uel.for`: 4271c66804e82df4b03692ea407c9197825d19488a0fc438efab28a2a537f51a
- Fixed reference input deck (`fixture_h0015.inp`): 9f3b52dca622a57eb0155b55e3474fc3108c90ebdb52c253dd658d51d9ad2a71
- Reconstructed nominal 1% input deck (`PK_M1_NOM1.inp`): 6385a4a49c4fa66c6b3e77f0a8276f7a62bc61e138a0a9d19451dcf878bb71ef
- Reference reaction force curve (`curve_standard_1398090.csv`): 780d05d39532e778802cdd8503cdd0df25568e412c3d4a51c50f4e001febede6
- Corrected nominal 1% curve (`curve_1404933_extracted.csv`): f71f01d150b12355f7a9f4a4a19563fe6e046ec31d94fe5abe85c103eadf3d56

Standard Abaqus CLI Reproduction Commands:

```

abaqus job=PK_M1_PRE input=PK_M1_PRE_CMS0020.inp interactive
abaqus cae noGUI=run_adaptive_remesh_canonical.py -- job=PK_M1_PRE
abaqus job=PK_M1_NOM1_STRICT input=PK_M1_NOM1_STRICT.inp \
    user=f42_mixed_uel.for cpus=1 interactive

```

References

- Abaqus 2023 Documentation: Analysis User's Guide*. Dassault Systèmes Simulia Corp., Providence, RI, USA, 2023.
- G. Molnár and A. Gravouil. 2d and 3d abaqus implementation of a robust staggered phase-field solution for modeling brittle fracture. *Finite Elements in Analysis and Design*, 130:27–38, 2017. doi: 10.1016/j.finel.2017.03.002.
- M. A. Msekh, J. M. Sargado, M. Jamshidian, P. M. Areias, and T. Rabczuk. Abaqus implementation of phase-field model for brittle fracture. *Computational Materials Science*, 96:472–484, 2015. doi: 10.1016/j.commatsci.2014.05.071.
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